

TELECOM CUSTOMER CHURN ANALYSIS

Methodology, Data Cleaning Process, and Analytical Findings

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Source file: telecom_customer_churn1.xlsx | 7,042 customer records | 38 fields

Total Customers	Churned	Joined	Stayed	Net Change
7,042	1,869	454	4,719	-1,415

1. Executive Summary

This document records the full analytical process applied to a telecom customer churn dataset of 7,042 customers across 38 fields. The work moves through four stages: a data cleaning and preparation audit, the selection and justification of metrics, the analytical findings for each business question, and the resulting recommendations.

The four business questions that framed the analysis:

1. How many customers joined the company during the last quarter?
2. What is the customer profile for a customer that churned, joined, and stayed? Are they different?
3. What seem to be the key drivers of customer churn?

Headline finding: The company experienced a net loss of 1,415 customers in the most recent quarter (454 joined vs. 1,869 churned). Month-to-month contracts and competitive pressure are the dominant churn drivers, with Competitor pressure accounting for 45% of all churns and month-to-month contracts associated with a 17x higher churn rate than two-year contracts.

2. Dataset Overview

The source file contains two sheets. The primary analysis sheet, **telecom_customer_churn**, holds one row per customer and was used exclusively for this work. A secondary pre-aggregated sheet (3 rows — count of Customer IDs, sum of Total Revenue, sum of Total Charges) was noted but not used in the analysis.

Property	Value
Total rows (customers)	7,042
Total columns (fields)	38
Total cells	267,596
Missing cells	30,844 (11.5% of all cells)
Columns with missing values	15 of 38
Grain	One row per unique customer

Fields in the dataset span five categories: **Demographics** (Age, Gender, Married, Number of Dependents, City); **Account details** (Contract, Tenure in Months, Payment Method, Paperless Billing, Offer); **Service subscriptions** (Phone, Internet, and 8 internet add-on services); **Financials** (Monthly Charge, Total Charges, Total Revenue, Total Refunds, Long Distance Charges, Extra Data Charges); and **Outcome fields** (Customer Status, Churn Category, Churn Reason).

3. Data Cleaning & Preparation

Before any analysis was run, every column was audited for missing values to distinguish genuine data-quality gaps from structurally meaningful blanks — cases where the absence of a value itself carries information (e.g., a blank in the 'Internet Type' field simply means the customer has no internet service).

3.1 Missing Value Audit

A column-by-column null count was run across all 38 fields. 15 columns contained missing values, totalling 30,844 missing cells. The table below groups them by the underlying cause of missingness, since the cause determines the correct treatment.

Column(s)	Missing	%	Root Cause
Churn Category, Churn Reason	5,173	73.5%	Only defined for churned customers — blank = 'Not Churned'
Offer	3,876	55.0%	Customer received no promotional offer
Internet Type, Avg Monthly GB Download + 8 internet add-on columns	1,526	21.7%	Customer has no internet service (Internet Service = No)
Multiple Lines, Avg Monthly Long Distance Charges	681	9.7%	Customer has no phone service (Phone Service = No)

3.2 Treatment Decisions

Churn Category & Churn Reason — conditional filtering

- **Decision:** Decision: Keep both columns. Fill with 'Not Churned' when analysing the full customer base. Filter to Customer Status = Churned (leaving both columns fully populated at 1,869 rows) when specifically analysing churn drivers.
- **Why:** Why: Dropping these columns would remove the only fields that record stated cause of churn — essential for answering Question 3. Dropping the rows would discard 73.5% of the dataset. Filtering at analysis time preserves both the full dataset and a 100%-complete churn subset.

Offer — fill with 'No Offer'

- **Decision:** Decision: Fill the 3,876 missing values with the category 'No Offer'.
- **Why:** Why: A missing Offer does not mean the data is unknown — it means no promotional offer was extended, a state that is itself meaningful and common (55% of customers). Treating it as its own category preserves this acquisition signal.

Internet Type, Avg Monthly GB Download, and 8 internet add-on columns — fill 'No Internet'

- **Decision:** Decision: Fill Internet Type with 'No Internet', Avg Monthly GB Download with 0, and all 8 add-on service columns with 'No Internet Service'.
- **Why:** Why: All 1,526 missing rows in this group correspond exactly to customers with Internet Service = No — confirmed by cross-tabulation. These are not data gaps; they are the expected absence of sub-services for customers who lack the parent service.

Multiple Lines & Avg Monthly Long Distance Charges — fill 'No Phone Service'

- **Decision:** Decision: Fill Multiple Lines with 'No Phone Service' and Avg Monthly Long Distance Charges with 0.
- **Why:** Why: All 681 missing rows correspond to Phone Service = No. A customer without phone service cannot have multiple lines or incur long-distance charges, so 0 / 'No Phone Service' is the factually correct fill, not an estimate.

3.3 Columns Dropped

No column was dropped from the working dataset. Every column with missing values was either filled with a logically correct category or handled through conditional filtering at the point of analysis. This approach was chosen over deletion because every missing-value pattern had a clear, verifiable structural explanation, and dropping affected rows would have removed up to 73.5% of the data.

Modeling note: The 8 internet add-on columns (Online Security, Online Backup, Device Protection, Premium Tech Support, Streaming TV, Streaming Movies, Streaming Music, Unlimited Data) are recommended for exclusion only from a predictive churn model's feature set, since their missingness is fully explained by Internet Service = No, making them redundant once that flag is included.

3.4 Post-Cleaning Validation

After applying fills, the dataset was re-checked to confirm zero remaining unexplained nulls outside the two conditionally-filtered churn columns. Cross-tabulations (Internet Type x Internet Service; Multiple Lines x Phone Service) were re-run to confirm every fill aligned 1:1 with its parent flag before any further analysis was performed.

4. Analytical Methodology & Metric Selection

This section documents the specific metrics chosen to answer each business question and the reasoning behind each choice. Metrics were selected to be directly interpretable by a non-technical business audience while remaining statistically sound.

4.1 Core Metrics Used Across the Analysis

Metric	Definition	Why This Metric Was Chosen
Customer Status segmentation	Customers grouped into Churned / Joined / Stayed	The dataset's native outcome label. Most direct, auditable way to answer 'how many joined/left' without building a derived proxy.
Tenure in Months	Months since the customer's account began	Used as a proxy for 'last quarter' (tenure ≤ 3 months) in the absence of an explicit join-date field, and as a loyalty indicator throughout.
Churn rate (%)	Churned customers $\div$ total customers in a segment	Raw counts mislead when segment sizes differ. Rate normalises for size, allowing fair comparison across contract types, internet types, etc.
Churn Category / Churn Reason frequency	Count and % share of each stated reason (churned customers only)	The only fields capturing stated cause rather than correlation — the most direct evidence for 'key drivers of churn.'
Cross-tabulation vs. churn rate	Two-way tables of account attributes $\times$ churn rate (%)	Surfaces structural risk factors (contract type, internet type, tech support) that show up in the data independently of stated reasons.

4.2 Question-Specific Metric Rationale

Q1 — Customers joined last quarter

- **Metrics used:** Filter Customer Status = 'Joined'; verify with Tenure in Months distribution; compute net customer change (joined minus churned); break down by Offer.
- **Rationale:** No explicit join-date column exists in the dataset. Tenure in Months ≤ 3 was validated as the 'last quarter' proxy by confirming that all 454 Joined customers had tenure of 1, 2, or 3 months — making the proxy internally consistent rather than assumed.

Q2 — Customer profiles by status

- **Metrics used:** Group-wise means for Age, Married %, Number of Dependents, Tenure, Monthly Charge, Total Revenue, Internet Service %, and Contract type distribution — computed separately for each of the three status groups.
- **Rationale:** A mix of demographic, financial, and contractual metrics was chosen deliberately so the profile captures who the customer is, what they pay, and how committed they are — the three dimensions a retention team would need to act on.

Q3 — Key drivers of churn

- **Metrics used:** Frequency distribution of Churn Category and Churn Reason (stated cause); churn rate (%) cross-tabulated against Contract type, Internet Type, Premium Tech Support, and Paperless Billing (structural cause).
- **Rationale:** Two complementary metric types were combined. Stated reasons show what customers say drove their decision; churn rate by segment shows what the data independently confirms — even for attributes the customer never mentioned. Churn rate (not raw counts) was used throughout to prevent large segments from appearing riskier simply because they contain more customers.

Key methodological choice: Churn rate (%) was used instead of raw churned counts whenever comparing segments of different sizes. This prevents large segments from appearing artificially riskier. For example, month-to-month customers are the largest contract group (3,610 of 7,042); their 45.8% churn rate is what demonstrates the risk — not their raw churned count.

5. Analysis Process & Findings

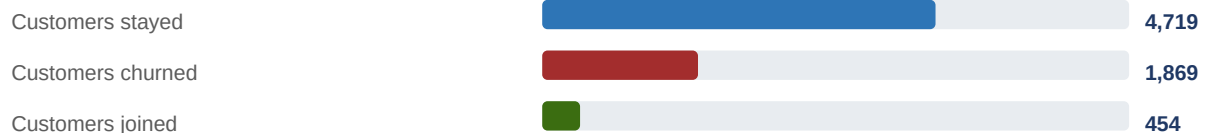
5.1 Question 1: How many customers joined during the last quarter?

Process

1. Filtered the cleaned dataset to Customer Status = 'Joined.'
2. Confirmed all Joined customers had Tenure in Months of 1, 2, or 3 — validating the last-quarter proxy.
3. Computed churned and stayed counts for the same period to establish net customer change.
4. Broke down the Joined group by Offer to identify which promotions, if any, drove new acquisition.

Findings

Metric	Value
Customers joined (tenure 1-3 months)	454
Customers churned (same period)	1,869
Customers stayed	4,719
Net customer change	-1,415
Joined customers receiving Offer E	175 (38.5%)
Joined customers with no offer	279 (61.5%)
Tenure split: 1 month / 2 months / 3 months	233 / 115 / 106



Insight: The company is in net customer decline — losing roughly 4 customers for every 1 gained. Offer E is the only promotion associated with new customer acquisition; all other offers (A-D) do not appear in the Joined group, suggesting they function as retention tools for existing customers rather than acquisition tools.

5.2 Question 2: What is the customer profile for churned, joined, and stayed customers?

Process

1. Grouped the dataset by Customer Status into three segments.
2. Computed mean Age, Married %, Number of Dependents, Tenure in Months, Monthly Charge, and Total Revenue for each segment.
3. Computed the % of each segment with Internet Service and on Month-to-Month contracts.
4. Compared all metrics side-by-side to test whether the three groups are practically distinct.

Findings

Attribute	Churned (1,869)	Joined (454)	Stayed (4,719)
Avg. age	49.7 yrs	42.9 yrs	45.6 yrs
Married	35.8%	18.5%	56.1%
Avg. dependents	0.12	0.36	0.62
Avg. tenure	18.0 mo	1.7 mo	41.0 mo
Avg. monthly charge	\$73.35	\$42.78	\$61.74
Avg. total revenue	\$1,971	\$120	\$3,736
Has internet service	94.0%	59.9%	73.9%
Month-to-month contract	88.6%	89.9%	32.8%

Insight: Yes — the three groups are strikingly distinct. Stayed customers are more likely married, have more dependents (family anchor), pay less per month, hold long-term contracts, and generate nearly double the lifetime revenue of churned customers (\$3,736 vs. \$1,971). Churned customers are mostly single, the oldest on average, pay the highest monthly charges (\$73.35), and are almost exclusively on month-to-month contracts. Joined customers are the youngest and lowest-spending group, consistent with an early-tenure cohort still evaluating the service.

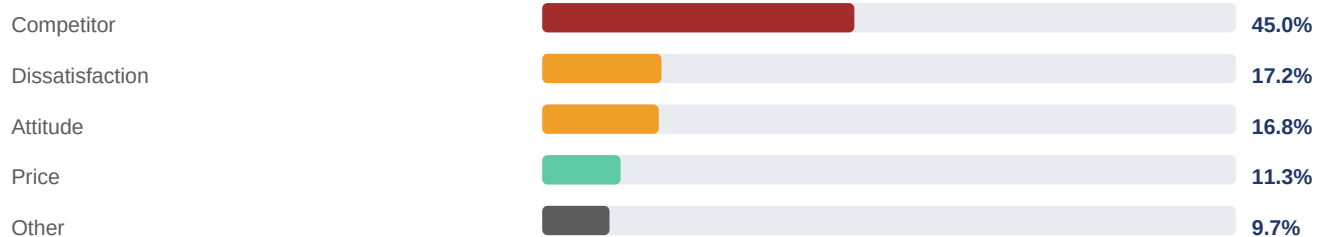
5.3 Question 3: What are the key drivers of customer churn?

Process

1. Filtered to Customer Status = 'Churned' (1,869 customers) — where Churn Category and Churn Reason are fully populated.
2. Computed frequency and % share of each Churn Category and the top 10 individual Churn Reasons.
3. Cross-tabulated churn rate (%) against Contract type, Internet Type, Premium Tech Support, and Paperless Billing.

Findings — Churn Category (stated reason)

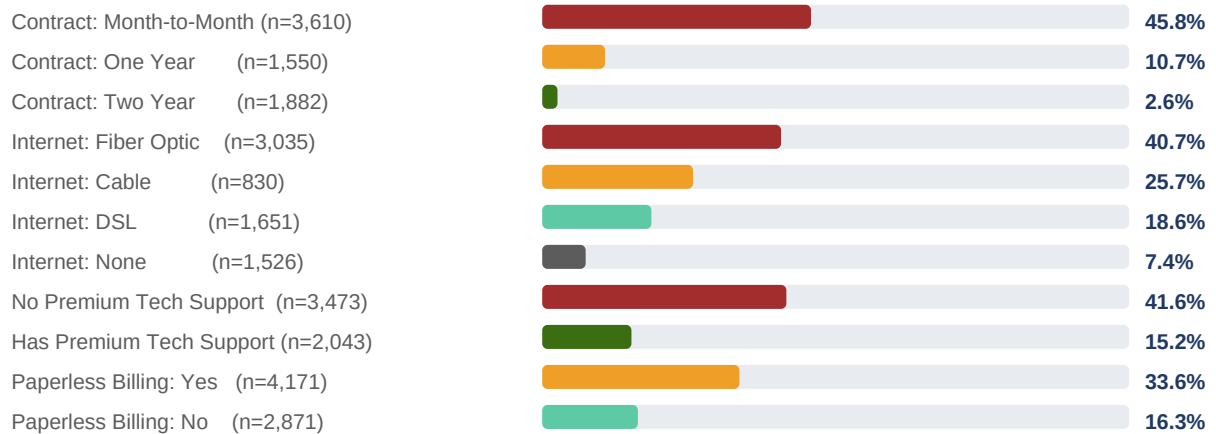
Churn Category	Count	% of Churned
Competitor	841	45.0%
Dissatisfaction	321	17.2%
Attitude	314	16.8%
Price	211	11.3%
Other	182	9.7%



Findings — Top 10 Specific Churn Reasons

Churn Reason	Count	% of Churned
Competitor had better devices	313	16.7%
Competitor made better offer	311	16.6%
Attitude of support person	220	11.8%
Don't know	130	7.0%
Competitor offered more data	117	6.3%
Competitor offered higher download speeds	100	5.4%
Attitude of service provider	94	5.0%
Price too high	78	4.2%
Product dissatisfaction	77	4.1%
Network reliability	72	3.9%

Findings — Structural Risk Factors (Churn Rate by Segment)



Insight: Competitive pressure is the single largest stated driver (45% of all churned customers), led specifically by device quality (313 cases) and better offers (311 cases). Contract type is the strongest structural predictor in the dataset — month-to-month customers churn at 45.8% vs. 2.6% for two-year contract holders (a 17x difference). Customers without Premium Tech Support churn at 41.6% vs. 15.2% with it. Fiber optic customers churn at 40.7%, suggesting a price-to-performance perception gap in that segment.

6. Recommendations

The following recommendations are derived directly from the metrics and findings in Section 5, prioritised by the size of the effect observed in the data.

1. Contract migration incentives — highest leverage

Moving a customer from month-to-month to a two-year contract is associated with a churn-rate reduction from 45.8% to 2.6% — the single largest effect in the analysis (17x difference). Since 88-90% of both churned and newly joined customers are on month-to-month plans, even a modest shift in contract mix would materially reduce churn. Offering meaningful incentives (discounted months, service upgrades) to migrate customers to longer terms is the most direct lever available.

2. Device upgrade programme for long-tenure customers

'Competitor had better devices' is the single most common specific churn reason (313 cases). A subsidised device upgrade path for customers past a tenure threshold (e.g., 24+ months) directly addresses the largest stated cause of loss.

3. Proactive, personalised retention offers

'Competitor made a better offer' is the second most common reason (311 cases). Offer E is the only promotion associated with new customer acquisition, while existing customers appear to receive offers only reactively. A proactive trigger for customers at high-risk junctures (e.g., approaching 18-24 months on a month-to-month plan) would engage them before a competitor does.

4. Premium Tech Support access and service quality

Customers without Premium Tech Support churn at 41.6% vs. 15.2% with it — nearly a 3x difference. Additionally, 'attitude of support person' was cited by 220 churned customers. This points to both a coverage gap (support not reaching enough customers) and a quality gap (the experience itself is driving departures).

7. Limitations & Next Steps

- **No explicit join/churn date field:** Tenure in Months was used as a proxy for recency. This is internally consistent but 'last quarter' is an approximation rather than a calendar-confirmed window.
- **Correlation, not causation:** Churn rate differences by segment (e.g., contract type, internet type) describe association, not proven cause. A controlled test — offering contract incentives to a randomised sample and measuring churn-rate change — would be needed to confirm causal impact before large-scale investment.
- **Self-reported churn reasons:** Churn Category and Churn Reason reflect what the customer (or a retention agent) recorded at cancellation, which can be incomplete or influenced by how the question was asked. The 130 'Don't know' responses (7.0% of churned customers) are a notable gap in stated-reason coverage.
- **Suggested next steps:** Build a predictive churn model (e.g., logistic regression or gradient boosting) using the cleaned feature set, excluding the 8 redundant internet add-on columns noted in Section 3.2. Run a small-scale pilot of the contract-migration incentive to validate the causal effect size before full rollout.

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